Geometry & Trigonometry

ANSWER KEY

Section 1: Angle Relationships & Polygons

1 In triangle ABC, the measure of angle A is 48° and the measure of angle B is 72°. What is the measure of the exterior angle at vertex C?

a. 110°

b. 120° ✓

c. 130°

d. 140°

The exterior angle theorem states that an exterior angle equals the sum of the two remote interior angles: $48^\circ + 72^\circ = 120^\circ$.

2 What is the sum of the interior angle measures of a regular octagon (8-sided polygon)?

a. 900°

b. $1,080^\circ$ ✓

c. $1,260^\circ$

d. $1,440^\circ$

Sum of interior angles is $(n-2) \times 180^\circ$. For $n = 8$: $(8-2) \times 180^\circ = 6 \times 180^\circ = 1,080^\circ$.

3 Two parallel lines are cut by a transversal. If one interior angle measures $(3x + 15)^\circ$ and its consecutive interior angle measures $(2x + 25)^\circ$, what is the value of x?

a. 24

b. 28 ✓

c. 32

d. 36

Consecutive interior angles are supplementary:
$(3x + 15) + (2x + 25) = 180 \Rightarrow 5x + 40 = 180 \Rightarrow 5x = 140 \Rightarrow x = 28$.

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Section 2: Right Triangle Trigonometry & Special Triangles

4 In right triangle ABC with right angle at C, $\sin A = \frac{5}{13}$. What is the value of $\cos B$?

a. $\frac{5}{13}$ ✓

b. $\frac{12}{13}$

c. $\frac{5}{12}$

d. $\frac{13}{5}$

In a right triangle, angles A and B are complementary ($A + B = 90^\circ$). By the cofunction identity, $\cos B = \sin A = \frac{5}{13}$.

5 If $\sin(2x - 10^\circ) = \cos(3x + 20^\circ)$, where all angles are acute, what is the value of x?

a. 14

b. 16 ✓

c. 18

d. 20

Since $\sin\alpha = \cos\beta \Rightarrow \alpha + \beta = 90^\circ$: $(2x - 10) + (3x + 20) = 90 \Rightarrow 5x + 10 = 90 \Rightarrow 5x = 80 \Rightarrow x = 16$.

6 A $30^\circ - 60^\circ - 90^\circ$ right triangle has a shorter leg of length 7. What is the area of the triangle?

a. $\frac{49\sqrt{3}}{2}$ ✓

b. $49\sqrt{3}$

c. 24.5

d. 98

Shorter leg $= 7$, longer leg $= 7\sqrt{3}$. Area $= \frac{1}{2} \times 7 \times 7\sqrt{3} = \frac{49\sqrt{3}}{2}$.

Section 3: Circles & Standard Equations

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7 What is the radius of the circle with equation $x^2 + y^2 - 6x + 8y - 11 = 0$?

a. 5

b. 6 ✓

c. 7

d. 8

Complete the square: $(x^2 - 6x + 9) + (y^2 + 8y + 16) = 11 + 9 + 16 = 36$. Therefore, $r^2 = 36 \Rightarrow r = 6$.

8 A circle in the xy-plane has center $(4, -2)$ and passes through the point $(7, 2)$. What is the equation of the circle?

a. $(x - 4)^2 + (y + 2)^2 = 25$ ✓

b. $(x + 4)^2 + (y - 2)^2 = 25$

c. $(x - 4)^2 + (y + 2)^2 = 5$

d. $(x - 7)^2 + (y - 2)^2 = 25$

Radius squared is distance squared: $r^2 = (7 - 4)^2 + (2 - (-2))^2 = 3^2 + 4^2 = 25$. Equation is $(x - 4)^2 + (y + 2)^2 = 25$.

9 In a circle of radius 10, a chord is drawn at a distance of 6 units from the center. What is the length of the chord?

a. 8

b. 12

c. 16 ✓

d. 18

The perpendicular from the center bisects the chord. By Pythagorean theorem on the half-chord: $h^2 + 6^2 = 10^2 \Rightarrow h^2 = 64 \Rightarrow h = 8$. Full chord length is $2 \times 8 = 16$.

Section 4: Arc Lengths, Sector Areas & Volume

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10 A circle has a radius of 12 cm. What is the length of an arc intercepted by a central angle of $\frac{2\pi}{3}$ radians?

a. 4π cm

b. 8π **cm** ✓

c. 12π cm

d. 16π cm

Arc length formula is $s = r\theta = 12 \times \frac{2\pi}{3} = 8\pi$ cm.

11 What is the area of a sector of a circle with radius 6 cm and central angle 60°?

a. 3π cm^2

b. 6π cm^2 ✓

c. 12π cm^2

d. 18π cm^2

Sector area formula: $A = \frac{\theta}{360^\circ} \times \pi r^2 = \frac{60}{360} \times \pi(6^2) = \frac{1}{6} \times 36\pi = 6\pi$ cm^2.

12 A right circular cylinder has a radius of 4 cm and a height of 9 cm. What is the volume of the cylinder?

a. 72π cm^3

b. 108π cm^3

c. 144π cm^3 ✓

d. 288π cm^3

Volume of cylinder: $V = \pi r^2 h = \pi(4^2)(9) = \pi(16)(9) = 144\pi$ cm^3.

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